Control Systems || 2024-2025's Final Exam (2nd Semester):

Question No. 1 (40 points):

Q1. For a given system, time response is $c(t) = 3e^{-7t} + 2$, the steady state output value is:

A) 5 B) 2 C) 3 D) Infinity

Q2. A unity feedback system with forward path $G(s) = \frac{1}{s^2+4s}$ is subjected to a unit step input. The Damping ratio of this system is:

A) 0.25 B) 0.5 C) 2 D) None

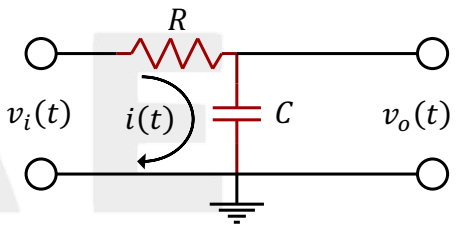
Q3. The unit impulse response of a first order system is $c(t) = 3e^{-0.25t}$. The time constant of the system is:

A) 4 seconds B) 5 second C) 0.25 second D) None

Q4. The unit impulse response of a first order system is $c(t) = 3e^{-0.5t}$. The transfer function of the system is:

A) $3/(s+0.5)$ B) $6/(s+1)$ C) $6/(s+0.5)$ D) None

Q5. For the system below, the output $V_o(s)$ is:



A) $I(s)/Cs$ B) $1/Cs$ C) $sI(s)/C$ D) None

Q6. For the system with a transfer function:

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K}$$

The system exhibits continuous sustained oscillations when subjected to a unit step input. What is the value of B?

A)1 B)0 C)0.707 D)None

■ For the root locus of a system with an open-loop transfer function:

$$G(s)H(s) = \frac{K(s+2)}{s(s+1)(s+4)}$$

Answer Q7, Q8, Q9, Q10, and Q11:

Q7. This system has a zero (undefined) at $s =$

A) $-1.5 + j\infty$ B) -0.55 C) $-1 + j\infty$ D) None

Q8. This system has one break-away point at $s =$

A) -1.5 B) -0.55 C) -1 D) None.

Q9. This system touches or intersects with the imaginary axis when $K =$

A) 0 B) 1 C) 6 D) None

Q10. This system is stable for:

A) $K \geq 0$ B) $6 > K > 0$ C) $K > 6$ D) None

Q11. There is a root locus on the real axis on the area:

A) $[-4, -2]$ and $[-1, 0]$ B) $[-4, -2]$ and $[0, \infty)$ C) $[-2, -1]$ and $[0, \infty)$ D)None

Q12. For a PID system with very high overshoot, an appropriate action may be:

A) Increase K_p B) Increase K_i C) Decrease K_d D) Decrease K_p

Q13. A system has the transfer function below, this system is:

$$\frac{C(s)}{R(s)} = \frac{1}{(s+3)(s+4)(s-1)}$$

A) Stable B) Not Stable C) Critically stable D) None

Q14. The Characteristic equation of a unity feedback system is $1 + GH(s) = 0$. Obtain the angle of $G(s)$:

A) $-90°$ B) $-180°$ C) $-270°$ D) $360°$

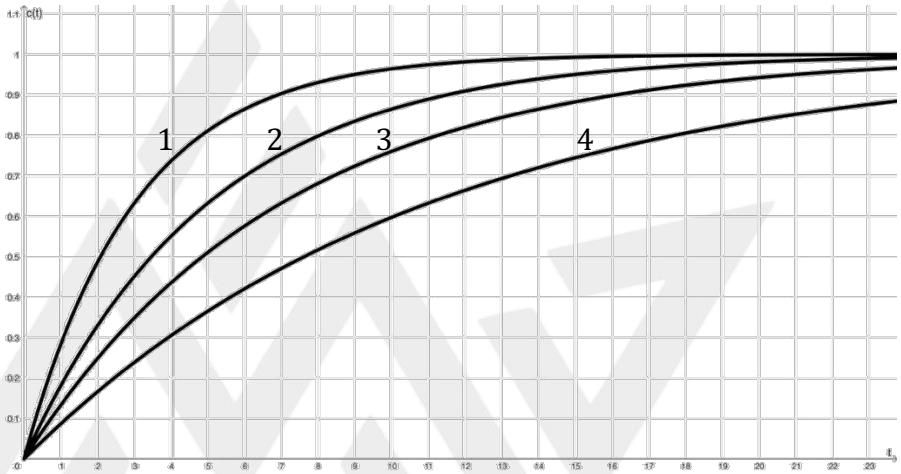
Q15. Curve number 3 is the time response of one of the first-order systems below:
(note: the input is 1)

A) $\frac{1}{3s+1}$

B) $\frac{1}{5s+1}$

C) $\frac{1}{7s+1}$

D) $\frac{1}{11s+1}$



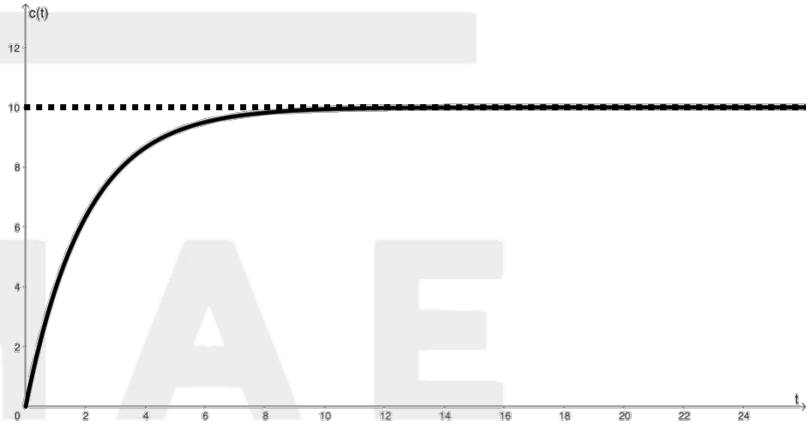
Q16. The time response of a controlled system is given. The used controller is:

A) P Controller

B) PI Controller

C) PD Controller

D) PID Controller



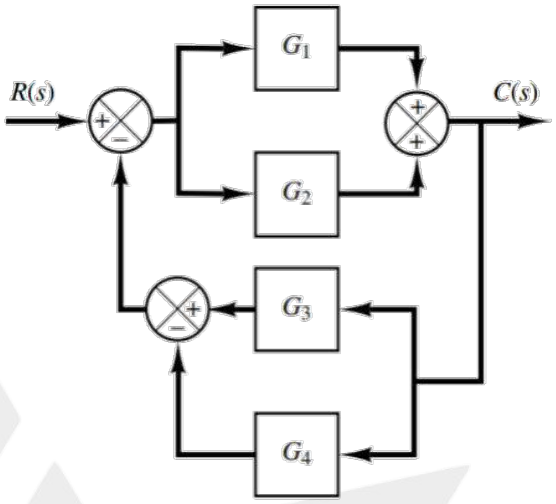
Q17. The characteristic equation of a given system is:

$$q(s) = s^4 + 2s^3 + (4 + K)s^2 + 9s + 25 = 0$$

This system is stable for:

A) $K > 0.5$ B) $K > 18/9$ C) $K > 0$ D) None

Q18. The transfer function of the system, is:



A) $\frac{1}{G_1G_2G_3G_4+1}$

B) $\frac{G_1G_2}{G_1G_2G_3G_4+1}$

C) $\frac{G_1+G_2}{(G_1+G_2)(G_3-G_4)+1}$

D) None

■ The transfer function for a given system (subjected to a unit step) is:

$$C(s) = \frac{K_1}{s^2 + K_1K_2s + K_1}$$

The system's overshoot is 25%. And undamped natural frequency is 1.72 rad/sec.

Answer **Q19 and Q20:**

Q19. The value of K_1 is:

A) 2.95 B) 0.404 C) 1.57 D) None

Q20. The value of K_2 is:

A) 2.95 B) 0.404 C) 0.471 D) None

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	C	A	A	A	B	A	B	A	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	D	B	B	C	B	D	C	A	C

Question No. 2 (10 points):

For a control system, the forward path and the backward path transfer functions are:

$$G(s) = \frac{10(s+1)}{(s+2)(s+3)}, \qquad H(s) = \frac{(s+3)}{(s+4)}$$

Obtain the error signal $E(s)$ and the steady-state error $e_{ss}(t)$. PROVIDE STEP-BY-STEP SOLUTION.

$$\frac{E(s)}{R(s)} = \frac{1}{1+GH(s)} \rightarrow E(s) = \frac{R(s)}{1+GH(s)} = \frac{R(s)}{1+\frac{10(s+1)(s+3)}{(s+2)(s+3)(s+4)}}$$

$$E(s) = R(s)\frac{(s+2)(s+4)}{(s+2)(s+4)+10(s+1)}$$

For the step input $\rightarrow R(s) = \frac{1}{s}$:

$$E(s) = \frac{1}{s}\cdot\frac{(s+2)(s+4)}{(s+2)(s+4)+10(s+1)} \rightarrow e_{ss}(t) = \lim_{s\to 0} sE(s) = \frac{(0+2)(0+4)}{(0+2)(0+4)+10(0+1)} = \frac{8}{18} = \frac{4}{9}$$

For the ramp input $\rightarrow R(s) = \frac{1}{s^2}$:

$$E(s) = \frac{1}{s^2}\cdot\frac{(s+2)(s+4)}{(s+2)(s+4)+10(s+1)} \rightarrow e_{ss}(t) = \lim_{s\to 0} sE(s) = \frac{1}{0}\cdot\frac{(0+2)(0+4)}{(0+2)(0+4)+10(0+1)} = \infty$$

For the parabolic input $\rightarrow R(s) = \frac{1}{s^3}$:

$$E(s) = \frac{1}{s^3}\cdot\frac{(s+2)(s+4)}{(s+2)(s+4)+10(s+1)} \rightarrow e_{ss}(t) = \lim_{s\to 0} sE(s) = \frac{1}{0^2}\cdot\frac{(0+2)(0+4)}{(0+2)(0+4)+10(0+1)} = \infty$$

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